Prior to discussing LASSO Regression, we must first discuss Ridge Regression. Hoerl and Kennard (1970) introduced ridge regression as a response to the instability of Ordinary Least Squares (OLS) estimation when predictor variables are in a condition known as multicollinearity. OLS estimates coefficients by minimizing the residual sum of squares, which can produce unstable coefficients when predictors are intercorrelated. What Ridge does different than OLS is that it adds an L2 penalty term to the OLS objective function as shown below:

Insert ridge regression equation

The penalty term adds a cost proportional to the squared size of each coefficient. The penalty term shrinks them toward zero as λ increases. When λ equals zero, ridge reduces to standard OLS. This penalty stabilizes coefficient estimates in the presence of multicollinearity by preventing any single coefficient from growing excessively large to compensate for correlated predictors. However, a key limitation of ridge is that it retains all predictors in the model. The coefficients do not reach zero which makes it less useful when the goal is to identify which variables matter most.

Tibshirani (1996) proposed the LASSO (Least Absolute Shrinkage and Selection Operator) which solves that limitation. While ridge uses an L2 penalty (squared coefficients), LASSO applies an L1 penalty using absolute values instead:

Insert LASSO regression equation

The L1 constraint region forms a diamond shape with sharp corners in coefficient space. When the optimization contours meet those corners, coefficients are driven to exactly zero. LASSO simultaneously shrinks coefficients and performs variable selection. Irrelevant predictors are completely excluded. Tibshirani's simulations demonstrated that LASSO performs best when there are a moderate number of relevant predictors. This is important because our goal with our paper is to see what predictors are most relevant in our dataset.

There are other methods related to LASSO that are worth mentioning. Hesterberg et al. (2008) provides a comprehensive review of Least Angle Regression (LARS), a method introduced by Efron et al. (2004) that unified LASSO, ridge, and forward stagewise regression. LARS works by incrementally adding variables to the model in the direction most correlated with the current residuals. LARS can actually be used to fit LASSO models Efron et al. (2004) and is generally pretty quick. While LARS provides insights into how penalized regression methods relate to one another, it is not employed in the present study as it is not needed. The current analysis prioritizes interpretability and direct variable selection.

LASSO has been tested in social science data numerous times. One example is it's applications in interpreting which familial factors effect father's well being the most. Ko et al. applied demonstration of OLS, ridge, LASSO, and elastic net regression using survey data from 277 U.S. fathers. Their findings demonstrated that while all three regularized approaches produced similar overall performance. They all did better than OLS by reducing overfitting, improving generalizability, and enhancing interpretability. LASSO was especially notable for shrinking less influential predictors to exactly zero in this study. Ko et al. found that the most relevant predictors (both positive and negative) were father involvement, parent-child relationship conflict, coparenting relationships, parental competency, work–family conflict, father role identity, and parent-child relationship closeness. Just as Ko et al. used LASSO to distinguish the most influential family factors from a broader set of correlated variables, the present study uses LASSO to identify which social media variables survive the longest under increasing penalty.